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Gamification in Education and Its Impact on Student Academic Performance: A Conceptual Model Based on Systematic Literature Review and PLS-SEM Analysis

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Abstract

Enhancing student academic performance remains a critical challenge for educators and administrators. Among various interventions, gamification has gained increasing attention as a promising approach to improving learning outcomes. However, existing research on the role of gamification in education remains fragmented due to its multidisciplinary nature. This study aims to synthesize current knowledge through a systematic review in order to develop and validate a conceptual framework linking gamification to student academic performance. By following the PRISMA framework, 62 relevant studies were reviewed, highlighting several recurring themes, including game design and development, student performance outcomes, and critical aspects of gamification such as cognitive development, motivation, and empowerment. Key mechanisms identified include active learning, personalized and adaptive learning, and collaborative interaction. Reported outcomes of gamification interventions include higher test scores, reduced anxiety and stress, and increased engagement and positive attitudes toward learning. Building on these findings, the resulting conceptual framework was validated through empirical research. Data were collected from 289 students from Saudi Arabia using a structured survey instrument, and Partial Least Squares Structural Equation Modeling (PLS-SEM) was employed for analysis. The results provide empirical support for the proposed framework, confirming gamification as a significant driver of improved student academic performance. The findings provide practical implications for educators and policymakers seeking to leverage gamification as a strategic tool for enhancing student learning outcomes.

Keywords: gamification; student academic performance; immersion; engagement; motivation

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1. Introduction

Education is universally recognized as a fundamental priority across societies [1]. Because of its critical role in social and economic development, key stakeholders—such as policymakers, educators, and institutions—are consistently engaged in efforts to improve the quality and accessibility of education [2]. This growing interest is mirrored in academia, where scholars from various fields increasingly examine educational issues [3]. Scholarly work has addressed topics such as pedagogy [4], curriculum development [5], teacher motivation [6], and educational infrastructure [7].

Moreover, student performance is a key and important issue that has attracted the attention of researchers [8]. The literature suggests that student performance, particularly

academic performance, has a varied yet significantly positive impact on key areas such as societal mental well-being [9], economic growth [10], and health [11]. Hence, improving student academic performance has become a key priority for educators, administrators, and policymakers [12]. Researchers have been developing and examining various interventions that can positively contribute to student academic performance. These interventions, as reported in the literature, include personalized tutoring [13], feedback [14], curriculum adjustment [5], mental health support [15], enhancement to the learning environment, and others.

Many interventions aimed at enhancing student performance are traditionally categorized as out-of-class interventions, which support learning beyond the classroom environment [16]. While such approaches can be valuable and effective, recent evidence indicates a growing interest among academicians, policymakers, and administrators in within-class interventions [17]. These are particularly significant as they allow educators to directly respond to each student's unique and personalized learning needs [18]. Within-class interventions enable teachers to adapt instructional methods, provide individualized feedback, and deliver targeted support with the goal of enhancing students' motivation [19], engagement [20], and knowledge acquisition [21]. By doing so, they can also help reduce disparities arising from differences in students' socio-economic backgrounds and cognitive abilities. The literature identifies several examples of effective within-class interventions, including differentiated instruction [22], formative assessments and feedback [23], and the integration of interactive technologies and tools [24].

As digital technologies have advanced and become more accessible, their use in educational settings has increased, especially in within-class interventions [24]. Existing literature has analyzed the use of various technologies, such as multimedia-equipped classrooms, remote meeting tools to conduct class activities, digital boards, and PowerPoint slides [24]. However, one key technique that can be effective is gamification, in which game-like elements are integrated into non-game contexts to increase motivation and engagement. According to [25], game methods play a significant role as within-class interventions that affect student academic performance. The use of gamification makes the learning experience more engaging and enjoyable, which increases student motivation [26]. This method improves academic achievement and retention by encouraging critical thinking, teamwork, and the application of knowledge in real-world situations [27]. Further empirical evidence suggests that gamification in the classroom directly enhances students' academic performance due to factors such as engagement and motivation [28]. Additionally, research shows that gamification helps students better understand challenging subjects such as anatomy, biology, chemistry, and math by making learning more immersive and enjoyable. This approach not only increases student involvement but also enhances their cognitive and critical thinking skills [29].

Since academic interests and research on gamification in education (particularly its role in enhancing students' academic performance) have increased, various key issues have emerged in existing literature. The key concern is to understand the current state of knowledge on gamification in classrooms and its impact on student academic performance. The literature is scattered, and empirical evidence regarding the relationship between gamification and academic performance is highly limited. Thus, synthesizing and summarizing the current state of knowledge and developing empirical evidence on gamification and student academic performance represent key gaps that need to be addressed. Therefore, the present research employed a systematic literature review (SLR) to collect, summarize, integrate, and synthesize the existing body of knowledge from different fields such as education, computer science, and information systems. Therefore, the current research aims to conduct an SLR to integrate interdisciplinary perspectives, understand the technological

landscape, and explore how such technology-driven gamification in classrooms facilitates student academic performance. Finally, the current study also aims to develop empirical evidence on the relationship between gamification and student academic performance. This evidence was generated by first constructing a conceptual model, derived from the SLR, which outlines the interplay among key relationships. Subsequently, primary data were collected using a survey questionnaire, and Partial Least Squares Structural Equation Modeling (PLS-SEM) was employed to validate the model.

Finally, the aim of this review is also to highlight significant implications for artificial intelligence (AI) and its integration with gamification. The role of AI in education has expanded across various aspects, particularly in enhancing student performance. The proposed conceptual framework offers important implications and insights for the design and execution of AI algorithms and models capable of determining and predicting student performance. More specifically, our framework facilitates the use of gamification and its key pedagogical facilitators (such as active learning) in AI models, which can collectively contribute to predicting student performance. Thus, integration of AI with gamification enables adaptive, personalized, and engaging learning environments by dynamically responding to learner behavior and performance. Overall, this framework provides a structured foundation for leveraging AI-driven gamification strategies to support effective and data-informed educational practices.

2. Literature Review

2.1. Gamification in Education

Gamification in education, particularly in the classroom, refers to integrating game elements (e.g., competitions and rules of play) into learning experiences to enhance students' learning and engagement. Gamification tends to make learning activities more enjoyable and immersive [27]. Gamification makes learning more engaging and rewarding by including features like stages, leaderboards, achievement badges, and point scoring into the instructional materials. This approach also makes learning more enjoyable and effective by appealing to students' innate drive for success and competitiveness [29]. Through interactive challenges and instant feedback, it improves retention rates, motivates students to take charge of their education, and develops a better comprehension of the subject matter [30].

2.2. Student Academic Performance

Student academic performance can be defined as the degree to which a student achieves educational goals and objectives, typically measured by quantitative indicators such as grades, test scores, and teachers' assessments [8]. Moreover, student academic performance encompasses the ability, aptitude, and attitude of the learner to understand, apply, and retain learning, and to apply this learning in practical settings [31]. Student academic performance is necessary for achieving various important educational objectives in general and student-related objectives in specific [32]. These objectives include motivation, health, long-term career success, and the development of talent for research and scientific advancement in society. Researchers have been evaluating various interventions that can help academicians improve student performance [33]. Gamification is an important intervention that academicians and students use to enhance their academic performance. Through immersion, engagement, and joyful experiences, gamification enhances the student learning experience by enhancing active and personalized learning [34].

3. Research Methodology

3.1. SLR Methodology

The existing literature outlines several methodologies for conducting an SLR [35–37]. Among these, one of the most widely recognized approaches is the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [36]. The PRISMA provides a structured process for identifying relevant literature, applying systematic inclusion and exclusion criteria, and synthesizing findings to generate theoretical insights and future research directions [38]. Following the precedent set by prior educational research [39,40], the present study adopts the PRISMA framework to ensure rigor and transparency in the review process.

3.1.1. Research Questions

The objective of the present study is to examine the relationship between educational gamification and student academic performance. Specifically, this research seeks to identify and analyze the key themes emerging from existing studies, with a particular focus on empirical research in this domain. It also aims to explore the main technologies applied in educational gamification and to assess their consequences for student performance. Finally, the study intends to provide recommendations for future research directions. To address these objectives, a set of research questions has been developed, as outlined in Table 1.

Table 1. List of Research Questions.

S.No.	Research Questions
1.	What are the key themes of research in gamification and student academic performance?
2.	What are the key technologies being used in gamification to enhance student academic performance?
3.	What are the key facilitators of student academic performance with respect to gamification?
4.	What are the key student academic performance factors that are targeted using gamification?

3.1.2. Database Selections

The literature database serves as a repository of studies targeted for the collection of literature to be reviewed and to answer research questions [41], as suggested by the PRISMA methodology. This research selected the databases of Scopus and Web of Science. These databases are widely recognized, high-quality sources of peer-reviewed research journals [42]. Therefore, both of these databases provide access to high-quality research publications. The choice of these databases is consistent with previous review studies conducted in the field of educational sciences [39].

3.1.3. Keywords for Searching Literature

Developing keywords is a critical step in conducting an SLR, as it enables researchers to identify and retrieve relevant studies from databases such as Scopus and Web of Science. This study follows the keyword development approach outlined by [43,44], as cited by [45]. According to these guidelines, keywords are derived from the major concepts of the review. For the present study, the core concepts are gamification and student academic performance. Accordingly, relevant key terms were generated to capture both dimensions, and Boolean operators (OR, AND) were applied to refine the search strategy. Table 2 presents the finalized set of keywords used in this research.

Table 2. Key word development.

S.No.	Keywords	Theme
1.	"Gamification" OR "game-based learning" OR "gameful design" OR "educational games" OR "digital game-based learning" OR "serious games"	Gamification
	AND	
2.	"student performance" OR "academic achievement" OR "educational outcomes" OR "cognitive development"	Student Academic Performance

3.1.4. Search and Results

This study used the keywords listed in Table 2 to search English-language research papers in Scopus and Web of Science, yielding 589 results (368 from Scopus, 221 from WoS). Python and its framework, such as Pandas and regular expressions, were used to remove duplicates by analyzing the titles of each publication. This process resulted in 399 unique papers. Each paper was screened for relevance using inclusion and exclusion criteria (see next section). The results can be seen in Figure 1.

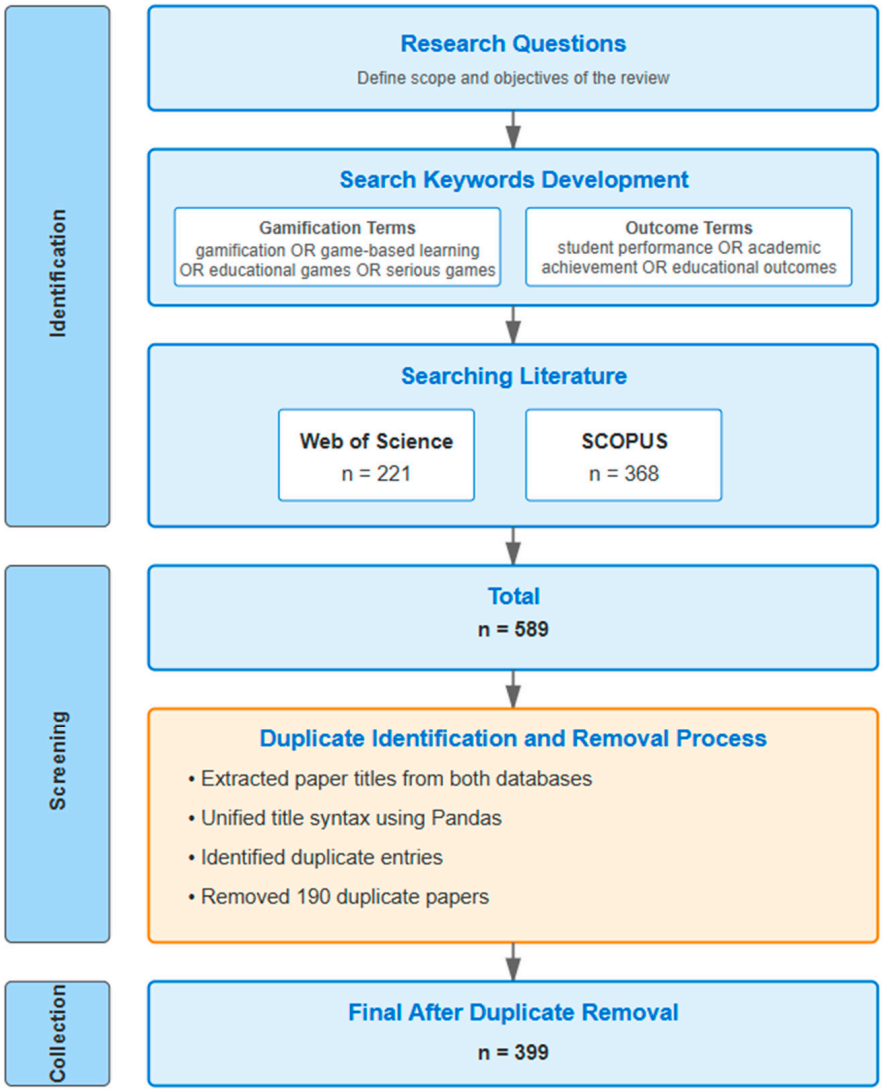

Figure 1. PRISMA framework search results.

3.1.5. Inclusion and Exclusion Criteria

The inclusion and exclusion criteria have played a key role in screening the publication data obtained after removing the duplicates. This research employed three key inclusion criteria. First, each publication addresses a key research gap at the intersection of gamification/games and education, with a focus on student academic performance. Second, the reviewed research papers were written in the English language and published in a recognized journal outlet. Thirdly, only research papers, both theoretical and empirical, were included in the final publication dataset.

Further, as far as the exclusion criteria are concerned, papers that do not focus on gamification and education were excluded from the review. The papers in which student academic performance, along with gamification, was not a variable were also excluded. Commentaries and other non-research publications were also excluded from the final review dataset.

These inclusion and exclusion criteria were applied by reading each abstract and introduction of the 399 research papers, which resulted in 62 final papers being included in the final review. Figure 2 presents these results.

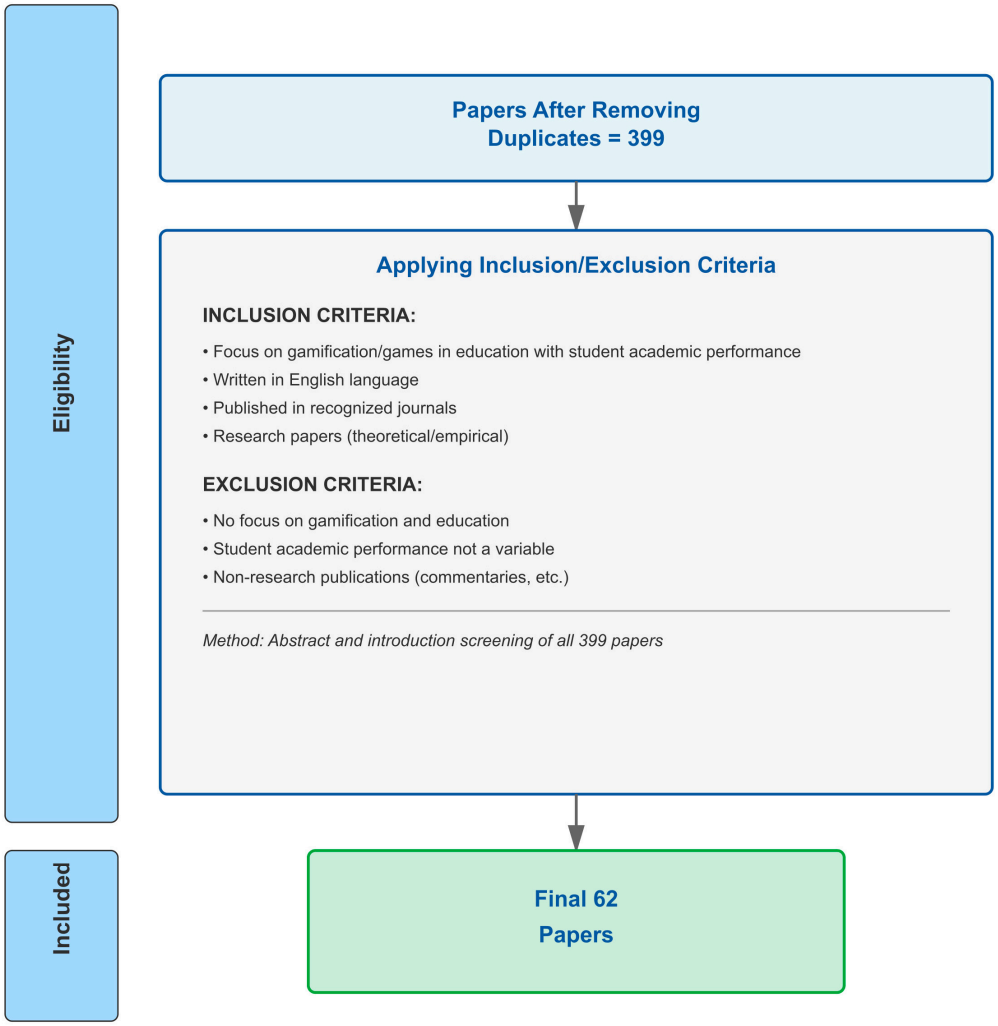

Figure 2. Applying inclusion and exclusion criteria.

3.1.6. Review Framework

This research developed a review framework that explicitly helps answer the research questions. The review framework attempted to code each paper in order to identify the relevant and dominant themes from the literature. The framework also includes the development of a list of technologies studied in the current research paper, the impact of gamification on student academic performance, the facilitating variables, and their consequence.

3.2. Empirical Methodology

The current study, following the development of a conceptual model as a result of SLR, has attempted to validate this model. The model validation using a quantitative research design helps us develop the empirical evidence on the nature of the relationship between the key factors in the model, including the role of gamification in facilitating learning activities that lead to student academic performance.

3.2.1. Data Collection Instrument

The present study employed a survey questionnaire as the data collection instrument, adapted from the existing literature. The data collection instrument was developed after the development of the conceptual model as a result of our SLR study. Table 3 presents key variables, measuring items, and their sources from existing literature. The five-point Likert scale was employed as the measuring scale in the data collection process. Appendix A presents the full data collection instrument.

Table 3. Instrument of data collection.

S.No.	Variable	No. of Items	References
1.	Game-Based Technology Adoption	4	[46]
2.	Multimedia Technology Adoption	4	[46]
3.	Digital Technology Adoption	6	[46]
4.	Augmented Reality	7	[47]
5.	Adaptive and Personalized Learning	14	[48,49]
6.	Collaboration and Intraction	6	[50]
7.	Active Learning	11	[51]
8.	Student Performance	19	[52]

3.2.2. Sample

This research collected data from students studying at different universities in Saudi Arabia. Saudi Arabia presents an interesting case, as the country has implemented and integrated digital technologies into its educational institutions under various initiatives. These technologies also include AR, VR, and gamification [53]. Therefore, the case of Saudi Arabia is an interesting and relevant context for developing empirical evidence with regard to gamification and academic performance.

Table 4 presents the descriptive statistics of the individuals who participated in the empirical study of the current research. The results show that the majority of participants are male (67.3%), with a considerable proportion of female (32.7%) participants. Moreover, the majority of participants are between 18 and 22 years old (44.3%), followed by those aged 22 to 27 years (28.5%). The age group between 28 and 32 years is also considerable (representing 22.3%). Furthermore, the key study disciplines of the participants were Information and Technology Science (29.5%), Business and Management (26.7%), and Medicine and Related Science (20.7%). Moreover, the majority of participants reported studying at either the undergraduate (48.5%) or graduate level (37.5%). Finally, consistent with the aim of the study, the majority of participants reported having employed gamification within the classroom and during the learning process.

Table 4. Participant demographics and study characteristics.

S.No.	Gender	Percentage
1	Male	67.3%
2	Female	32.7%
S.No.	**Age**	**Percentage**
1	18–22	44.3%
2	22–27	28.5%
3	28–32	22.3%
4	33 and more	4.9%
S.No.	**Field of Study**	**Percentage**
1	Business and Management	26.7%
2	Information Technology Science	29.5%
3	Engineering	16.5%
4	Medicine and related science	20.7%
5	Arts, Social Sciences, and Humanities	6.6%
S.No.	**Level of Study**	**Percentage**
1	Higher School	9%
2	Undergraduate	48.5%
3	Graduate	35.7%
4	Post-Graduate	6.8%
S.No.	**Use of Gamification in Class Teaching**	**Percentage**
1	Yes	73.7%
2	No	26.3%

3.2.3. Analysis of Data

To validate the conceptual model that will lead to the development of empirical research, the study has employed PLS-SEM. PLS-SEM is a multilevel regression technique that helps test the direct and indirect relationships between variables, enabling the development of robust empirical evidence. The present research employs the measurement and structural models of PLS-SEM. The measurement model helps us to assess reliability, validity, and whether the data are suitable for the research model, while the structural model statistically assesses the relationships among different variables [54].

4. Findings

This section presents the results of both SLR and PLS-SEM analyses. The SLR results provide an in-depth review of the existing body of literature, which serves as the foundation for developing the proposed conceptual model. In turn, the PLS-SEM results offer a comprehensive evaluation that empirically validates the conceptual model. Together, these two approaches not only highlight the theoretical underpinnings of the research but also provide robust statistical evidence that strengthens the credibility and practical relevance of the findings.

4.1. Analysis of SLR Descriptive Data

4.1.1. Publications Timelines

The current study has undertaken an analysis of the publication timeline. The results show that the first publication on the topic of gamification and its impact on student academic performance appeared in the year 2002 [55]. Research on gamification and student academic performance began to take effective shape in 2013. Following 2013, a drastic increase in research to understand the impact of gamification on student academic performance has been observed. The results further show that the year 2023 has seen the highest number of papers assessing the impact of gamification on student academic performance. The results are shown in Figure 3.

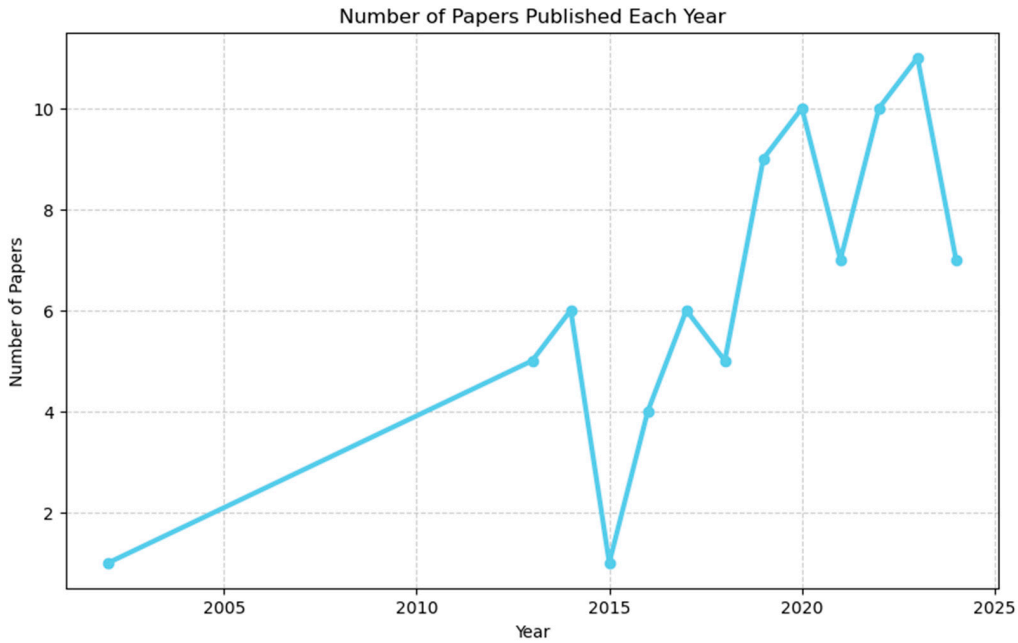

Figure 3. Publication Timeline.

4.1.2. Top Journals

This study has also assessed the top journals that are accepting and publishing research on gamification and student academic performance. According to the results on top journals (see Figure 4), a higher number (six in total) of papers on gamification and student academic performance are published by the *Education and Information Technologies* journal. Following *Education and Information Technologies*, other important journals publishing research on education, such as *Advances in Physiology Education*, *Frontiers in Psychology*, and the *International Journal of Engineering Education*, have published the most studies. These journals are also important and widely recognized journals in the field of educational science. Finally, other journals, mostly related to educational science, such as *Computer and Education*, *Journal of Research in Science Teaching*, and the *International Journal of Gaming and Computer-mediated Simulations*, have also published a significant number of studies.

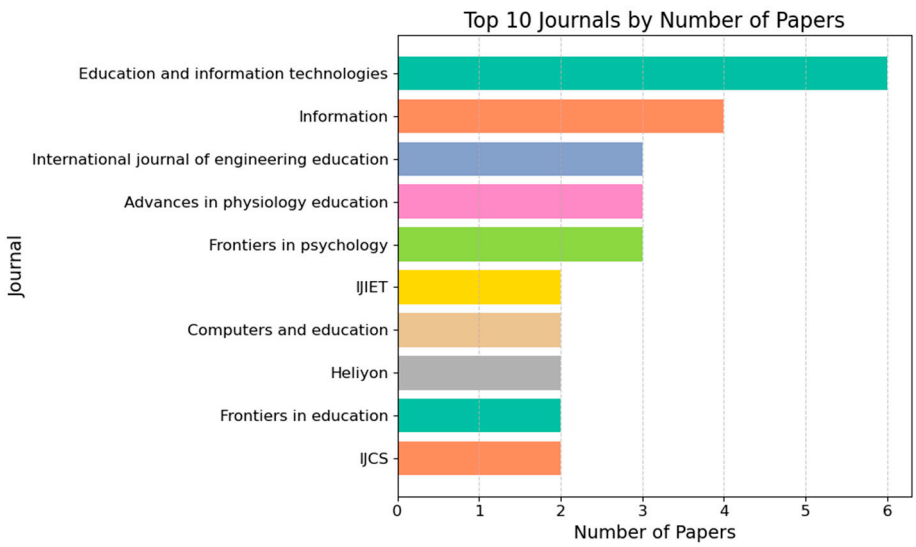

Figure 4. Top journals. (IJIET = *International Journal of Information and Education Technology*. IJCS = *International Journal of Gaming and Computer-Mediated Simulations*).

4.1.3. Top Authors

The research has also attempted to yield the results of the most prolific authors in the area of gamification and student academic performance. The results suggest that various researchers, as mentioned in Figure 5, have taken a keen interest in gamification and student academic performance. According to the findings, these authors have each published two articles on the relationship between gamification and academic performance and achievement. Thus, it can be argued that the 20 papers out of 62 were published by the 10 authors, suggesting that the area of gamification and student academic performance is becoming an increasingly important research field. The researchers are continuously exploring new research gaps and working to fill them.

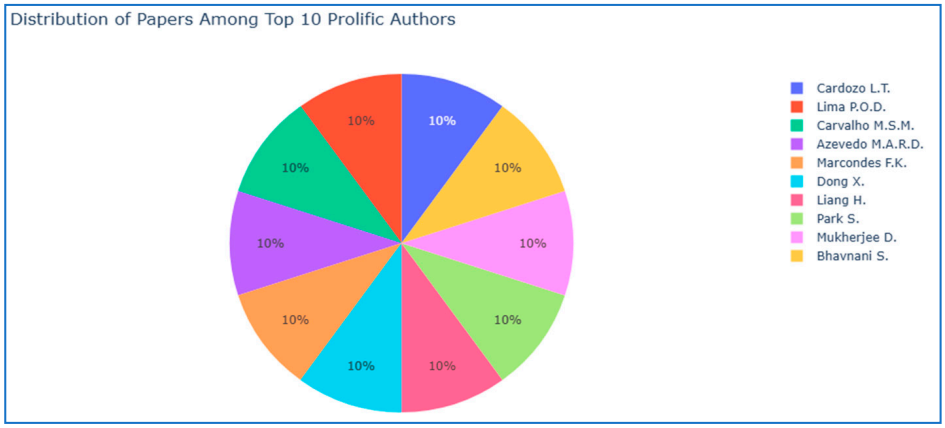

Figure 5. Prolific authors.

4.1.4. Country Heat Map and Top Countries

This study has also attempted to yield the results of the top countries with respect to the authors' affiliations. The results of the top countries have also been visualized using a country heatmap. The results of the top countries and country heatmap can be seen in Figures 6 and 7, respectively. The purpose of such analysis is to understand the scope of the current and existing research geographically and culturally. It also helps us to suggest further areas for future research and make recommendations to include a more culturally and geographically diverse sample.

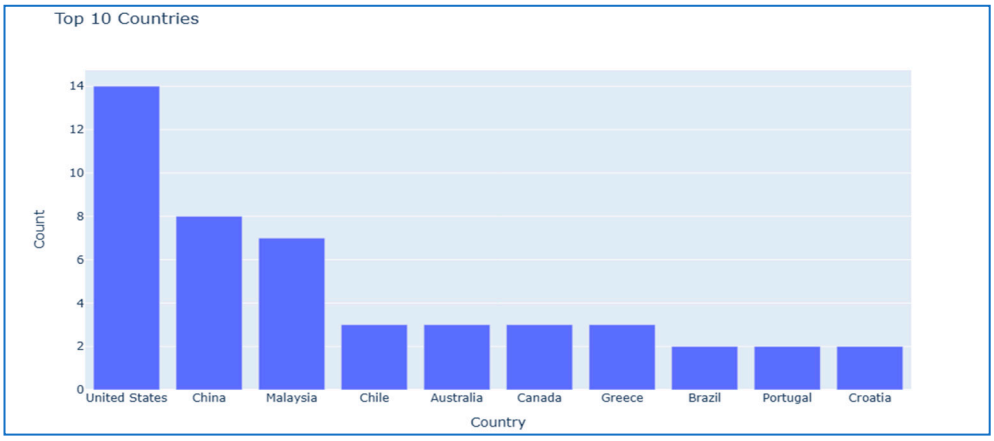

Figure 6. Top countries.

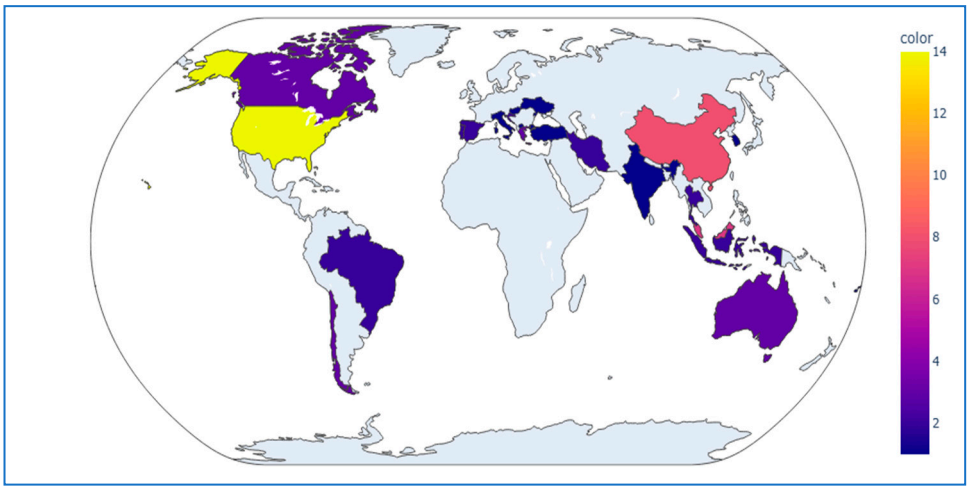

Figure 7. Country heatmap.

The results of the top countries with respect to authors' affiliations suggest that the United States of America is leading in publishing research related to gamification and student academic performance. The United States of America (USA) is followed by China in second place and Malaysia in third place. Other countries, such as Australia, Canada, and Greece, have also conducted a significant amount of research. Finally, researchers from other countries, including Brazil, Portugal, Croatia, India, and Iran, have also taken an interest and published in the area of gamification and student academic performance.

Following Figures 6 and 7 shows the country heatmap, which shows that research is mostly dominant in both industrialized countries like the United States of America and China, and in newly industrialized countries, including Malaysia, Australia, and others.

4.1.5. Key Findings from Review

The present research has also attempted to summarize the existing literature using thematic analysis by comprehensively analyzing the published studies (see Table 5). The results suggest that researchers (in the context of education) have attempted to focus on the nature, design, and testing of the gamification itself. The interest in gamification is driven by various questions regarding different key aspects, such as pedagogical development and its effectiveness in the classroom settings. Following the growing interest in gamification, the results suggest that the existing literature has comprehensively focused on the empirical aspects of the outcomes of gamification used as within-class interventions. The most significant outcome of gamification as an educational intervention is student academic performance. From the results, it can be inferred that the interest and focus on the design of gamification are contingent upon the fact that researchers, administrators, and policymakers are interested in understanding the various outcomes related to students. Besides student academic performance, these outcomes also include motivation, cognitive development, and empowerment.

Furthermore, the most recent use of gamification as a within-classroom intervention is based on the extensive use of technology. The present research has attempted to extract and synthesize the knowledge with regard to the technological landscape of gamification. The results suggest that various technologies are playing an important role in designing and introducing gamification as in-class interventions. These technologies include game-based technologies such as interactive animations, mascots, and avatars. Moreover, other technologies, such as augmented reality, multimedia, and digital technologies, are also among the key tools used for the development and execution of gamification.

Table 5. The results of the review framework.

S.No.	Variable	No of Items	References
1	Themes	Gamification	[56–74]
		Student and Educational Performance	[75–93]
		Cognitive Development	[94–96]
		Motivation	[97–105]
		Student Empowerment	[97–105]
2	Technologies	Game-Based Technology	[67,80,84,94,97,100,104,106–108]
		Multi-Media Technology	[101,103,109]
		Augmented Reality	[58,63,89,104]
		Digital Technologies	[76,84,87,96,109,110]
3	Performance Facilitators	Active Learning	[59,97,111]
		Personalized and Adaptive Learning	[57,100,106]
		Collaboration and Interactions	[75,98,112]
4	Key Performance Indicators	Higher Test Scores	[56,75,106]
		Reduced Levels of Student Anxiety and Stress	[87,97,101]
		Higher Levels of Student Engagement and Attitude	[73,113]

Additionally, the research attempted to identify the key factors that can facilitate student academic performance, including high scores on tests and academic exams. These facilitating factors include active learning, personalized and adaptive learning, and collaboration and interactions. The results suggest that technology-driven gamification enhances test scores through these facilitating factors. It also helps students to reduce anxiety and stress levels while encouraging higher levels of student engagement and attitudes. Table 5 summarizes the key findings from the review of the literature.

4.2. The Themes of Gamification

This research, using the review framework, has attempted to develop dominant themes discussed in the selected research papers. The themes were extracted from the literature by coding each paper based on the key topics and research problems addressed. The results suggest that five dominant themes have emerged from the literature. These themes are discussed in the following sections.

4.2.1. Gamification

The review reveals that one of the key dominant themes being discussed in the current literature pertains to gamification itself [56–61]. Recent studies highlight how gamification is transforming teaching and learning [65–67]. The literature emphasizes core aspects of gamified education, particularly innovations in teaching methods and student learning outcomes [68–70,114]. Interactive and serious games have notably improved the teaching of complex science subjects such as chemistry and biology [60,67,71]. Likewise, gamified approaches have enhanced student learning, especially in challenging areas like mathematics [58,63,64,72]. Moreover, emerging technologies such as augmented reality, multimedia, and digital platforms are reshaping the development of gamification [57,73,74].

4.2.2. Student and Educational Performance

Based on an extensive literature review, this research concludes that the second most dominant theme in the literature pertains to student and educational performance [75–80]. Student performance includes various key aspects such as learning and development [81], participation in class activities [80], and curricular and extracurricular activities [82]. The literature suggests that gamification brings a high level of student interactivity and engagement with educational content, particularly course material [81,83–89,115]. Such engagement with course material positively impacts active participation in class, including

in discussions of topics and during experiments [75,80,82,90–92]. Finally, the literature also suggests that gamification, by increasing engagement, can have a positive impact on students' active participation in both curricular and extracurricular activities [81,84,85,88,93].

4.2.3. Cognitive Development

The results of the current review suggest that a key outcome of gamification is the cognitive development of the students. The research concludes that there is a positive impact of gamification on cognition and cognitive development [94]. It has been suggested that gamification, through its engagement and interactivity, has significantly contributed to the development and improvement of working memory. The research argues that the extensive use of gamification for educational purposes may help students to improve working memory and skill development [95]. Thus, gamification as an intervention can play an important role in enhancing memory. The literature suggests that memory is a critical part of cognitive abilities, and improvement in working memory will have a significant positive effect on the cognitive development of the students [94]. Finally, an interesting insight regarding skill development has also been discussed in the literature. It has been highlighted that gamification positively impacts working memory, which can result in the development of various skills, such as computing and mathematics [96].

4.2.4. Motivation

The fourth most important theme synthesized from the review relates to student motivation. The current study concludes that increasing student motivation is a key aspect of current research focusing on utilizing gamification in academic settings [97–100]. Motivation is referred to as a key element of students' behavior. It has been suggested that a higher level of motivation results in a higher level of student academic and non-academic performance [101–104]. Motivation becomes a key factor that helps students achieve higher scores in various subjects, as well as learn and master tougher subjects, including mathematics, chemistry, and biology [97,104,105]. This research has found a positive impact of gamification on student motivation. Engagement, interactivity, and entertainment in gamified learning activities have a positive impact on motivation. These three factors (engagement, interactivity, and entertainment) impact in a way that they simplify and make various key concepts easier for the students. Thus, learning and mastering such key concepts enhances the student's motivation [99,100].

4.3. Key Technologies Being Employed in Gamification

This research has conducted a review of the literature to understand the key technologies used in the gamification of education. Technology plays a key role in the development and innovation of games and gamification. Technology is also making it possible to make games accessible on a wider scale. Further, technology aims to further improve interactivity and engagement within games. The present research's review literature has identified four key sets of technologies that have been employed in education (See Table 6). The following section presents an in-depth analysis of each technology.

Table 6. Key technologies used for games.

S.No.	Key Technologies	Components
1.	Game-Based Technologies	Artificial intelligence, UX/UI, C#, audio processing devices, and physics modeling
2.	Multimedia Technology	Graphics, audio–video, and visuals
3.	Augmented Reality	AR headsets, software, and others
4.	Digital technology	Multimedia

4.3.1. Game-Based Technology

Game-based technology refers to a combination of elements (e.g., computer software and hardware) that enables the creation and playing of games for educational purposes [67,84,97]. These games can be both simulation and serious games [94,104]. Game-based technologies play an important role in the development of games for various purposes [67,80]. According to an estimate, the game has evolved into a separate technology industry, with revenue projected to reach SAR 300 billion [107]. Various key technologies, such as User Experience and Interface (UX/UI), AI, programming languages, and other hardware components, such as audio processing devices and physics simulations, play an important role in game development. While the broader literature addresses game-based technology from both end-user and back-end perspectives, AI's role extends beyond mere technical infrastructure. It fundamentally reshapes pedagogical interactions through adaptive algorithms, intelligent tutoring systems, and personalized learning pathways. The integration of AI within educational gaming represents a convergence of computational intelligence and instructional design, enabling real-time responsiveness to individual learner needs, dynamic difficulty adjustment, and data-driven insights into student progress patterns [94,100,106].

4.3.2. Multimedia Technology

The second key technology discussed in the current literature regarding gamification in education is known as multimedia technology. Multimedia technology provides the much-needed interactivity and engagement to students through its captivating graphics, visuals, videography, and audio [109]. The aspect of interactivity has been made possible with the help of captivating visuals, graphics, and other key components of multimedia technology [103]. The literature suggests that multimedia technology has changed the way games are conceptualized and has positively impacted the end-user (student) satisfaction [101].

4.3.3. Augmented Reality

Augmented reality (AR) is referred to as a technology that provides an immersive view of digital objects in the real-world environment [104]. Research on gamification suggests that AR is the next big innovation within the gaming industry and gamification [89]. The present review suggests that academics have been employing AR-driven games in educational settings. AR-driven games provide immense opportunities for academics to introduce a high-level of innovation in teaching various key subjects, such as biology and physics [63]. Furthermore, AR-driven technology embedded in games can also provide an opportunity for students to learn such subjects in a highly interactive manner [58].

4.3.4. Digital Technology

This research has synthesized the important role of digital and internet technologies in gamification and student performance. The current study's results suggest that digital and internet technologies have been playing a key role in the development and provision of gamification [84]. Digital and internet technology has enhanced the quality of multimedia and video games supported by smart devices [110]. It has also helped end users experience the games with significant interactivity and engagement [76]. Furthermore, the literature also suggests that combining digital and internet technology with other technologies like AR and multimedia can further enhance the experience of games. Finally, new gaming platforms have also been introduced with the help of digital and internet technology. These new gaming platforms have impacted games by enabling various users to contribute to game development and play these games. Thus, in the educational setting, game platforms

have enabled both students and academicians to cooperate in game development and improvement [87,96,109].

4.4. Key Learning Facilitators

The current study has also attempted to understand the key factors that can facilitate students' academic performance with the help of gamification in education. The results of an in-depth literature review suggest that there are three factors that determine students' academic performance in educational gamification. These factors include active learning, personalized and adaptive learning, and collaboration and interactions.

Active learning is a process in which students immerse themselves in the learning process by actively participating in learning activities. The results suggest that gamification actively promotes active learning. Simulative and serious games require students to participate in the key activities, which entail decision-making. Thus, such levels of participation and decision-making result in active participation in the gamification activities. Furthermore, such active participation results in students immersing themselves in a game situation, leading to learning various aspects and elements. Thus, the final outcome of such active learning contributes to the students' academic performance [59,97,111].

Secondly, the results found that gamification also promotes personalized and adaptive learning. Adaptive and personalized learning refers to tailoring learning material to individual students' needs and acumen. The results also suggest that gamification actively promotes adaptive and personalized learning. The games actively help academicians design and execute them according to the aptitude, acumen, and learning needs of individual students. Thus, the ability of academicians to design games and participation according to individual student learning needs promotes adaptive and personalized learning. Hence, such personalized and adaptive learning contributes to positive academic performance [57,100,106].

Thirdly, the results of the present review study have also suggested that social interaction and collaboration powered by games contribute to positive academic performance. The results indicate that interaction and collaboration are basic features of games. Gamification promotes collaboration and interactivity among the participants and academics. Such levels of participation and interactivity promote active collaboration among students and academicians. Collaboration helps students exchange knowledge among themselves and learn actively from each other, which ultimately enhances and improves their academic performance [75,98,112].

4.5. Key Performance Indicators

The results of the SLR revealed that gamification can help to enhance key student performance outcomes. The results suggest that there are three factors that influence student academic performance in educational gamification. These factors include students achieving higher test scores, reduced levels of student anxiety and stress, and higher levels of student engagement and attitude.

Gamification in education has been associated with higher cognitive ability among learners. It has been argued that, due to the active learning of students using the games, students are better able to learn important concepts. Thus, it will also help students to score higher in examinations and relevant tests [56,75,106]. Secondly, the pressure of exams, learning new concepts, and achieving relevant academic performance has always been linked with stress and anxiety, which have various detrimental effects on mental health. It has been found that gamification and its impact on student academic performance can have a significant effect on the reduction in stress and anxiety [87,97,101]. Finally, higher academic performance with the help of gamification in education can improve student

confidence. Students with a good level of academic performance tend to be more confident and have an attitude that encourages critical thinking. Thus, it can be argued that higher student academic performance due to gamification can imbue students with confidence and a positive attitude in the long term [73,113].

4.6. Conceptual Framework and Validation

Based on SLR, this study also attempts to build a conceptual framework that highlights the interplay of the relationship of gamification, its key technology, facilitating factors, and key consequences. The conceptual framework, as shown in Figure 8, illustrates that technologies such as multimedia tools, augmented reality, game-based technology, and other digital technologies are key technologies used in the design, development, and execution of gamification. These technologies enable gamification to produce dynamic learning environments that go beyond conventional methods of teaching. The framework specifically emphasizes three major facilitators enhanced by gamification, which include collaboration and interaction, personalized and adaptable learning, and active learning. Finally, the framework highlights the relationship between these facilitators and the key consequences of technology-enabled gamification. These consequences include higher exam performance, reduced levels of anxiety, and higher level of engagement. The empirical validation of the framework will fulfill an important research gap, along with other key studies that need to be undertaken in the future.

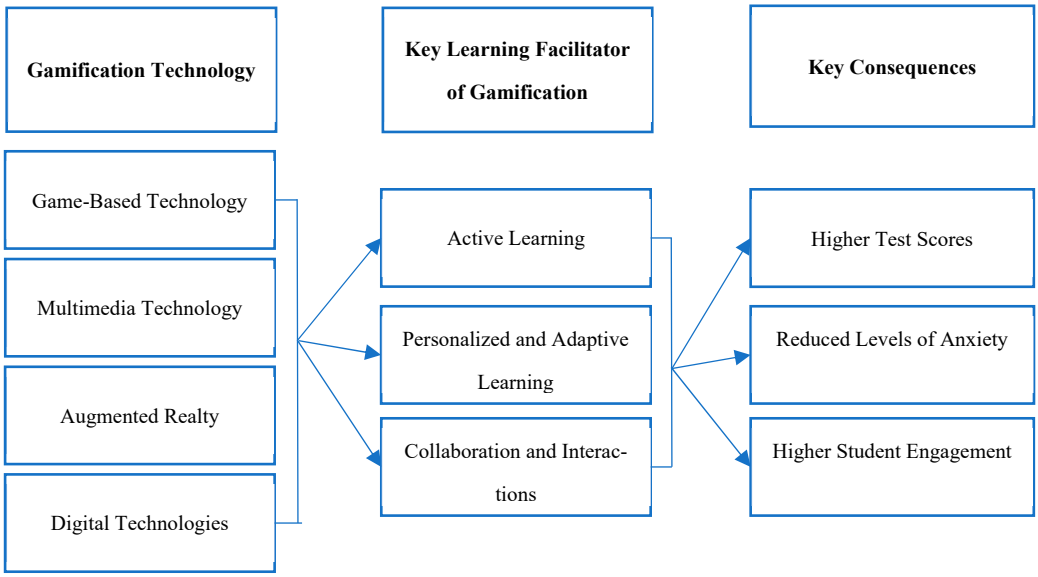

Figure 8. Conceptual framework.

The conceptual model developed in this study provides important insights into the key variables that explain how gamification, both as a technology and as a pedagogical methodology, can enhance student performance in classroom settings. While these theoretical contributions are significant, such insights may be limited due to the absence of empirical validation. To address this gap, the present research integrates quantitative evidence by employing a structured survey instrument and analyzing the data through PLS-SEM.

4.6.1. Hypothesis and Testing Model Digital Technology

The conceptual model, as shown in Figure 8, provides a graphical illustration of the key constructs and variables, as well as their relationship. The following section provides details on the conceptualization of each construct.

Gamification Technology

The results from the systematic review have helped us to understand that gamification in the educational context results from introducing certain technologies as part of either classroom activities or the curriculum [116]. Thus, based on such an assertion, this research hypothesizes that gamification in education is supported by various key technologies that promote active learning among students [116]. The findings from the literature have also yielded four distinct types of such game-based technologies that influence learning facilitation and student academic performance.

First, game-based technologies are digital tools and platforms that use game design elements (like points, levels, challenges, and rewards) to engage users in non-gaming contexts such as education, training, or behavior change [117]. Therefore, it can be concluded that these technologies have emerged and are being dominantly used.

Second, multimedia technology is a broad term that, in educational contexts, primarily refers to learning management systems (LMSs) such as Blackboard Mobile or similar platforms, which are designed to promote and facilitate active learning [118] and enhance overall student performance [119]. In the context of gamification, these technologies integrate interactive multimedia elements—including graphics, audio, video, and animation—to create immersive learning experiences that boost student engagement and motivation. Such game-integrated multimedia systems enable educators to deliver content dynamically, provide immediate feedback, and track learner progress, thereby transforming traditional passive learning into an active, participatory process.

Thirdly, the present research's SLR findings illustrate the important role of digital technology in educational settings, which tends to enhance student performance by facilitating collaboration among students and various stakeholders, including teachers and peers [120]. In the realm of educational gaming, digital technologies provide collaborative platforms where students can engage in multiplayer learning activities, share strategies, and solve problems collectively within game-based environments. These digital tools support both synchronous and asynchronous communication by enabling real-time teamwork during gameplay and fostering social learning through discussion forums, shared leaderboards, and cooperative challenges. This ultimately creates a more connected and interactive educational ecosystem.

Finally, the current SLR has highlighted an increasing role of augmented reality as both a technology and an educational intervention, along with gamification [119]. The empirical literature illustrates that augmented reality (both generally and within educational gamification) tends to promote adaptive and personalized learning, thereby improving students' academic performance [121].

Key Learning Facilitator of Gamification

The results of the current SLR have pointed out that gamification can facilitate learning in three different forms.

First, active learning—which our review found to be enhanced using gamification and related technologies—promotes interactive and engaging experiences that encourage student participation and exploration. Game features, including points, badges, levels, and instant feedback, are aspects that motivate students to take ownership of their learning journey [59].

Second, a key learning form facilitated by gamification and technologies is personalized and adaptive learning. Such a form provides tailored educational experiences that respond dynamically to each student's unique abilities, pace, and learning style [99].

Finally, collaboration and interaction (facilitated by gamification technologies) are achieved by creating dynamic learning environments where students actively engage with

peers, instructors, and educational content through shared game-based experiences. These platforms incorporate multiplayer features, team challenges, and cooperative problem-solving tasks that require students to communicate, negotiate strategies, and pool their knowledge to achieve common goals within the game context [98].

Key Outcomes

The SLR study has helped us to conclude that the key outcome of gamification as an intervention is student academic performance, which is further distinguished into three key interconnected outcomes. First, higher test scores (self-reported) reflect the direct cognitive gains students achieve through gamified learning [81]. Second, reduced anxiety emerges as students experience learning in a low-stakes, supportive environment where failure is reframed as part of the learning process rather than a punitive outcome [84]. Third, higher levels of engagement manifest as students become intrinsically motivated participants in their learning journey. Game elements such as rewards, progression systems, and narrative contexts capture attention, sustain interest over time, and encourage active participation that extends beyond mandatory requirements [85]. In the present research's empirical part, performance was conceptualized as perceived student academic performance. The conceptualization of performance as perceived academic performance is consistent with both the methodology and findings of the SLR. The perceived academic performance measurement using the employed scale helped us to self-report the performance in terms of test scores and other key aspects, such as anxiety and engagement. Further, as aspects (e.g., high test scores) may require different methods, self-reported scoring can be effective from a theoretical and methodological standpoint.

In conclusion, this research's empirical findings (in contrast to the SLR) suggest that a key consequence of gamification and other variables is student academic performance, which is measured using [52] and incorporates various key dimensions, as illustrated in the conceptual framework in Figure 8.

4.6.2. Measuring the Constructs

The present study has employed a survey instrument to measure the study's key constructs. The key technology constructs (i.e., game-based technology, multimedia technology, and digital technology) were measured using a technology adaptation and propensity scale [46]. Apart from that, the augmented reality construct was measured using its acceptance scale developed by [47]. Further, key facilitating variables such as adaptive and personalized learning were measured using a scale developed by both [48,49]. Similarly, collaborative and interactive learning were measured using the scale developed by [50], while active learning was measured using the methods of [51]. Finally, student academic performance was measured holistically using the [52] scale. The scale by [52] is considered a robust and well-established instrument that captures academic performance from multiple dimensions, including self-reported examination scores, student engagement levels, and reduced anxiety [122]. Thus, measuring student academic performance holistically is consistent with the theorization proposed in this study.

4.6.3. Common Method Bias

To address various key concerns associated with survey research, such as the validity of causal inference and interpretation, this research has undertaken the test of the Harman Single Factor Test [123]. Table 7 presents the results of Harman's Single Factor Test to assess common method bias, indicating that it is not a significant concern. The first factor accounts for only 13.794% of the total variance, which is well below the conservative threshold of 50%, suggesting that no single factor dominates the variance in the dataset.

Table 7. Harman Single Factor Test.

Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	9.829	14.892	14.892	9.104	13.794	13.794

4.6.4. Measurement Model

The present research, consistent with the guidance in PLS-SEM literature, has assessed the measurement model [54]. The measurement helps assess the reliability and validity of the constructs [124]. Such reliability and validity assessments of the constructs ensure that each construct is measured correctly and effectively, and that the data also reports a proper fit to the regression model used to assess the hypothesis and coefficient in the relationship. Thus, establishing an effective measurement model is a prerequisite for the structural model [54,124].

Table 8 shows the key results of the measurement model, i.e., construct reliability and validity. Cronbach's alpha and composite reliability tend to measure reliability by assessing the inter-item consistency of the items within each construct. The acceptable threshold for both Cronbach's alpha and composite reliability is 0.70 [124]. The earlier results show construct validity and reliability for certain constructs below 0.70. Thus, to address this issue, this study referred to the outer loading results, where items with factor loadings less than 0.70 [124] were removed. By doing so, the factor loading results achieved the minimum threshold values, as shown in Table 8. The average variance extracted (AVE) assesses construct validity by assessing the accuracy of each item in measuring its intended constructs. The literature suggests that an AVE of 0.50 can be assumed to indicate that a particular construct effectively demonstrates validity [124]. The results in Table 8 suggest that all constructs report AVE above 0.5.

Table 8. Construct reliability and convergent validity.

Constructs	Cronbach's Alpha	Composite Reliability	Average Variance Extracted (AVE)
Active Learning	0.976	0.979	0.809
Adaptive and Personal Learning	0.973	0.976	0.744
Augmented Reality	0.943	0.959	0.853
Collaboration	0.947	0.957	0.789
Digital Tech	0.920	0.943	0.805
Game-Based Tech	0.943	0.959	0.854
Multimedia Tech	0.950	0.964	0.868
Performance	0.979	0.981	0.732

Apart from the AVE, the validity of constructs is also measured through discriminant validity [125], which assesses how each construct is being measured differently from other constructs in the PLS-SEM model. In addition, the Heterotrait–Monotrait Ratio of Correlations (HTMT) is a key statistical test used to measure discriminant validity in the PLS-SEM model. Table 9 presents the results of the HTMT, and according to the literature of PLS-SEM, the HTMT value for a construct should be below the threshold of 0.80 [126]. The results suggest that almost all constructs report an HTMT value lower than 0.80, and it can be concluded that discriminant validity is assumed effectively.

The regression model (PLS-SEM in this study) derives its explanatory and predictive power from the proportion of variance in the endogenous variables accounted for by the exogenous variables. This variance explanation is typically assessed using the coefficient of determination (R-squared, R^2). The R-squared value indicates how well independent constructs predict the dependent constructs. The results of the R-squared are presented in Table 10, which shows that the final dependent variable, student perfor-

mance, exhibits a substantial variance explanation of 73.8%. In addition, the endogenous constructs—active learning (27.4%), adaptive and personal learning (18.6%), and collaboration (14.5%)—also show meaningful levels of explained variance, reflecting moderate explanatory power. Collectively, these findings confirm that the model captures a significant portion of the variance in key constructs, thereby underscoring the robustness and validity of the proposed framework.

Table 9. Discriminant validity.

	AL	APL	AR	C	DT	GBT	MT
Adaptive and Personal Learning (APL)	0.063						
Augmented Reality (AR)	0.151	0.444					
Collaboration (C_)	0.058	0.602	0.126				
Digital Technology (DT)	0.070	0.139	0.050	0.394			
Game-Based Technology (GBT)	0.542	0.087	0.186	0.082	0.138		
Multimedia Technology (MT)	0.096	0.061	0.083	0.355	0.086	0.137	
Performance (P)	0.491	0.651	0.142	0.609	0.201	0.248	0.139

Table 10. Variance explanation using R-squared.

S.No.	R-Squared	R-Squared—Adjusted
Active Learning	0.274	0.268
Adaptive and Personal Learning	0.186	0.182
Collaboration	0.145	0.141
Performance	0.738	0.734

Finally, the current study assessed the overall model fit using the Standardized Root Mean Square Residual (SRMR) test in PLS-SEM. Table 11 presents the SRMR results, revealing a saturated model value of 0.048 and an estimated model value of 0.132. The saturated model SRMR of 0.048 is well below the recommended threshold of 0.08 [124], indicating an excellent fit when all possible relationships are considered. However, the estimated model SRMR of 0.132 exceeds the acceptable threshold, suggesting some discrepancy between the observed correlations and those implied by the theoretical model. This divergence between saturated and estimated models is not uncommon in complex models [124].

Table 11. Model Fitness.

	Saturated Model	Estimated Model
SRMR	0.048	0.132

4.6.5. Structural Model

Finally, this research has assessed the structural model to assess the statistical relationship between constructs and estimate the coefficient. The bootstrapping procedure, involving the creation of 5000 subsamples, is used to estimate the structural model. The results of the structural model are presented in Table 12, as shown in Figure 9. The results show that all relationships are accepted based on a significant value (p-value) of less than 0.05, except for the relationship between multimedia technology and active learning, which is not significant (p-value equals 0.736).

Table 12. Structural path coefficients and hypothesis testing results.

Hypothesis	Coefficient	Standard Deviation	T Statistics	*p* Values
Active Learning -> Performance	0.508	0.043	11.872	0.000
Adaptive and Personal Learning -> Performance	0.481	0.038	12.749	0.000
Augmented Reality -> Adaptive and Personal Learning	0.431	0.053	8.061	0.000
Collaboration -> Performance	0.316	0.042	7.604	0.000
Digital Tech -> Collaboration	0.381	0.049	7.790	0.000
Game-Based Tech -> Active Learning	0.521	0.052	10.061	0.000
Multimedia Tech -> Active Learning	−0.017	0.051	0.337	0.736

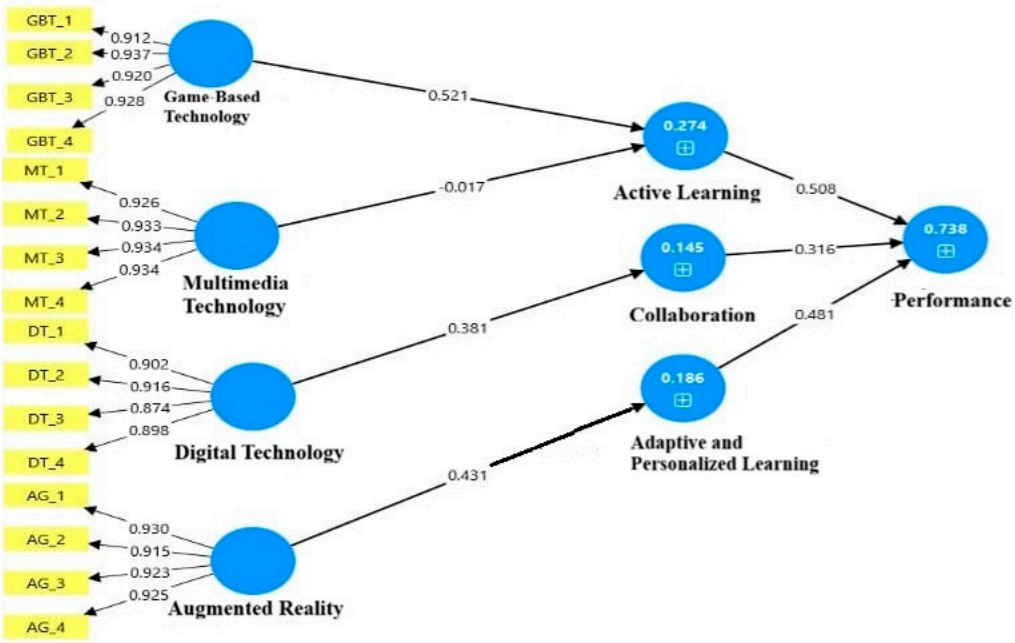

Figure 9. PLS-SEM path model.

5. Discussion

The present research adopts an integrated two-stage method, involving an SLR and conceptual model validation using a survey questionnaire. In the following section, the findings for both methods are discussed.

The SLR findings have revealed both an interesting and multifaceted landscape of gamification in the education context, where technology integration is key to introducing gamification as an intervention. The research findings resulted in five significant and distinct themes of research that are being explored in this area. These themes include gamification and its design, key outcomes such as student performance, cognitive development, motivation, and student empowerment. However, the literature is dominated by game applications, with empirical attention primarily focused on the key student performance indicators. Thus, it can be concluded that most current (especially empirical) studies illustrate the key benefits and policy implications of gamification. Further, the results indicate factors such as motivation, cognitive development, and student empowerment as distinct themes. These findings indicate a paradigm shift from teacher-centered to learner-centered approaches, where gamification serves not merely as a delivery mechanism but as a catalyst for intrinsic engagement and learner agency.

Apart from the themes, the technological landscape represents another key aspect of the SLR findings. The review results highlight the strategic role of key technologies, including game-based technologies, which have attracted the most substantial research attention. In this regard, AI emerges as a pivotal component that not only enhances game mechanics and user experiences but also enables adaptive learning systems capable of dynamically

responding to individual student needs. Additional research is also needed to further examine, empirically and theoretically, the emerging technology of AI in gamification. Thus, it can be argued that specific tools and techniques positioned within gamification, as both a process and educational intervention, have generated considerable interest among researchers. Such tools, along with technologies such as augmented reality and multimedia technologies, represent an evolution toward immersive and multimodal learning experiences. Finally, regarding performance, the current findings are divided into two key aspects, including facilitators and outcomes. Performance facilitators represent an interesting aspect in which gamification (as an intervention) tends to play a role in facilitating learning. Thus, key aspects such as active learning, personalized adaptive learning, and collaborative interactions appear consistently across the literature. These key learning facilitators of gamification serve as theoretical bridges between technological implementation and measurable outcomes. Finally, the current SLR has identified three key performance outcomes or indicators that educational administrators and researchers are focused on regarding gamification as an educational intervention. These outcomes include improved test scores, reduced anxiety and stress, and enhanced engagement. These outcomes reveal that contemporary educational technology research employs a holistic evaluation framework extending beyond traditional academic metrics to encompass affective and behavioral dimensions of learning effectiveness.

The present research has attempted to empirically validate the theoretical model using PLS-SEM as the analytical strategy. The research hypothesized that gamification and the introduction of gamification as an educational intervention, enabled by various technologies, can facilitate learning and lead to optimal student academic performance. The empirical findings provide strong support for the hypothesis being tested. The results have shown that there exists a positive and significant relationship between game-based technology and active learning (p = 0.000, β = 0.521) [117]. These results suggest that game-based technologies serve as powerful facilitators of active learning, transforming students from passive recipients of information into engaged participants who actively construct knowledge through interactive gameplay experiences. Furthermore, the data reveal that this game-facilitated active learning exerts a significant positive influence on student academic performance (p = 0.000, β = 0.508) [127]. This indicates that when students are actively involved in their learning process through gamified environments, they achieve measurably better academic outcomes [117].

Furthermore, the results revealed an interesting relationship between effective and efficient use of digital technologies and collaboration among students. The results (β = 0.381, p = 0.000) suggest that technology-enabled platforms effectively facilitate peer interaction, knowledge sharing, and cooperative problem-solving in educational settings. Subsequently, this enhanced collaboration significantly contributes to improved student academic performance (β = 0.316, t = 7.604, p = 0.000). This suggests that when students actively engage with peers through digitally mediated collaborative activities, they achieve better learning outcomes. This two-stage relationship underscores the critical role of digital technologies not merely as direct performance enhancers, but as enablers of social learning processes that create collaborative environments where students benefit from collective intelligence [120,127].

Finally, the findings of this research have also revealed an interesting, significant, and positive effect of augmented reality on adaptive and personalized learning. The results indicate (β = 0.431, t = 8.061, p = 0.000) that augmented reality technologies effectively create customized learning experiences that respond to individual students' needs, preferences, and learning pace by overlaying digital content onto physical environments in contextually relevant ways. Subsequently, this enhanced adaptive and personalized learning

significantly contributes to improved student academic performance (β = 0.481, t = 12.749, p = 0.000), suggesting that when students receive tailored instruction that adapts to their unique learning trajectories through AR-enabled environments, they achieve substantially better academic outcomes [121].

Further, an important aspect of the current research is its implication for designers of AI models and algorithms. Recently, AI has become a key factor in both predicting and enhancing student academic performance. However, any AI model, from simple regression to neural networks, needs to work based on certain inputs to generate an output, including prediction. This research provides a significant implication with regard to designing algorithms that can help to detect students' academic performance. These include active learning, collaboration, and personalized learning. An important aspect is measuring these aspects and converting them into data that can be provided as input for training. However, as the role of synthetic data is increasing, survey data that effectively measure these variables can be used to train such models. Overall, it can be concluded that this research offers significant implications for AI models by providing a framework to detect and predict student academic performance.

Overall, this empirical study provides robust evidence that gamification, augmented reality, and digital technologies can significantly enhance student performance when these key factors are integrated as part of within-class interventions. The empirical findings confirmed that the use of gamification tends to foster active learning, and such active learning enhances students' academic performance. Similarly, digital technologies enhance such performance by promoting collaboration among students in different ways. Finally, the role of augmented reality in supporting adaptive and personalized learning was also validated, highlighting its potential as a transformative educational tool. Collectively, these results underscore the value of technology-enhanced pedagogies in creating engaging, collaborative, and effective learning environments. Beyond theoretical validation, the study offers practical implications for educators and policymakers seeking to design innovative strategies that maximize student performance in technology-driven education systems.

6. Conclusions

Gamification, as both an educational intervention and an integral part of the curriculum, presents a significant and transformative opportunity to innovate the education system and enhance its provision as a public good. The current research (both SLR and empirical survey) found that gamification, along with its related technologies, such as digital platforms and augmented reality, can be considered a key educational intervention. It first significantly facilitates the learning process and ultimately leads to student academic performance. The research concludes that three interconnected mechanisms collectively transform educational experiences and outcomes of positive student academic performance. First, game-based technologies serve as powerful catalysts for active learning by transforming students from passive recipients of information into engaged participants who actively construct knowledge through interactive gameplay experiences. Second, augmented reality technologies emerge as particularly effective enablers of personalized and adaptive learning by creating customized educational experiences that dynamically respond to individual students' needs, preferences, and learning pace. Third, digital technologies facilitate critical collaborative learning processes by creating platforms where students actively engage with peers through shared problem-solving, knowledge exchange, and cooperative challenges.

Such facilitation and improvements in the learning process have empirically and theoretically been found to enhance student academic performance. These facilitators can play a role as input features for AI-based systems, enabling the systematic analysis of the student academic performance indicators. By leveraging these inputs, AI models can

effectively detect, assess, and predict student academic performance, thereby supporting timely interventions and personalized learning strategies. From an empirical standpoint, the current study demonstrates that students engaged in gamified learning environments report consistently higher self-reported scores (confirmed through empirical evidence). They also experience reduced anxiety and stress levels, along with higher engagement, as proposed by SLR findings. These aspects of performance are attributed to the supportive, low-stakes nature of game-based learning, where failure becomes an opportunity for growth rather than a source of fear. These comprehensive performance gains collectively validate gamification as a holistic educational approach that enhances not merely what students learn, but how they experience and relate to the learning process itself.

Beyond these immediate outcomes, this research establishes a robust foundation for educators, administrators, and policymakers to confidently integrate gamification into mainstream educational practice. It encourages them to view gamification not as a peripheral innovation but as an evidence-based pedagogical strategy with demonstrated effectiveness across diverse learning contexts. As educational systems worldwide confront challenges of engagement, equity, and relevance in an increasingly digital world, technology-enhanced gamification emerges as a compelling solution that bridges traditional pedagogical wisdom with contemporary technological capabilities, ultimately contributing to more accessible, effective, and engaging education for all learners.

Limitations

The present research also has some key limitations that need to be addressed in future research. First, the research has attempted to build empirical evidence on the framework using the survey method; however, experimental and longitudinal studies can further enhance the findings and strengthen the evidence. Secondly, the research framework can be used in a qualitative study, such as semi-structured interviews, to further strengthen and improve the framework. Third, in the SLR study, gamification and educational games were considered and reviewed. Although both concepts are similar and consistent, some conceptual differences exist, and this review may have led to some conceptual overlap despite their theoretical distinctions. Thus, inclusion of studies on educational (video) games (which are few) in SLR may have broadened the scope beyond gamification alone. Therefore, future studies should clearly distinguish and independently analyze gamification and educational games to improve conceptual clarity and strengthen theoretical precision.

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Data Availability Statement: The data related to SLR and empirical analysis can be made available upon request.

Conflicts of Interest: The author declares no conflicts of interest.

Appendix A

Table A1. Instrument.

		Item Statement	Outer Loading
Game-Based Technology	1	I can figure out new game-based technologies or applications without help from others.	0.912
	2	I seem to have fewer problems than others when using game-based technologies.	0.937
	3	People often come to me for advice on using new game-based technologies.	0.920
	4	I enjoy figuring out how to use new game-based technologies or platforms.	0.928

Table A1. *Cont.*

		Item Statement	Outer Loading
Multimedia Technology	1	Multimedia technologies help me perform tasks at the time and in the way I prefer.	0.926
	2	Multimedia technologies help me make positive changes in my personal or professional life.	0.933
	3	New multimedia tools make my daily routines easier and more efficient.	0.934
	4	Using multimedia technologies (like video, audio, and interactive media) gives me more control over how I manage my day.	0.934
Digital Technology	1	Sometimes I feel that digital technology controls my life more than I control it.	0.902
	2	I feel overly dependent on digital technology in my everyday life.	0.916
	3	The more I use digital technologies, the more I feel tied or bound to them.	0.874
	4	I need to be cautious while using digital technologies, as they could expose me to criminal activity.	0.898
Augmented reality	1	I find augmented reality useful in my daily life.	0.930
	2	Using augmented reality increases my chances of achieving things that are important to me.	0.915
	3	Using augmented reality helps me finish things faster.	0.923
	4	Using augmented reality increases my productivity.	0.925
Adaptive and personalized learning	1	The teacher sets personalized learning goals for me based on my learning data.	0.872
	2	The teacher designs personalized learning tasks based on my learning data to help me better understand course topics.	0.897
	3	The teacher provides me with challenging tasks suited to my learning level through the smart platform.	0.898
	4	The teacher provides feedback on my progress toward achieving learning goals through the smart platform.	0.913
	5	The teacher monitors my learning progress in real time and adjusts teaching content and methods accordingly.	0.886
	6	After completing learning tasks, the smart platform provides detailed error analysis and suggestions for improvement.	0.879
	7	The smart platform recommends suitable learning tasks to help me achieve my learning goals.	0.865
	8	The teacher uses smart tools to recommend diverse learning materials (videos, articles, exercises) for me.	0.886
	9	The smart platform generates a personalized list of learning resources to help me complete learning tasks more effectively.	0.840
	10	The smart platform automatically analyzes my learning weaknesses based on my data (quiz results, study time, etc.).	0.840
	11	When I encounter learning difficulties, the smart platform recommends suitable learning resources or solutions.	0.859
	12	The teacher provides multiple learning content options, and I can choose the most suitable content based on my interests and needs.	0.800
	13	I can participate in the course in different ways (online discussions, virtual experiments, group collaboration) through the smart platform.	0.792
	14	The teacher offers various ways to showcase learning outcomes (text, charts, videos) allowing me to choose the best method.	0.835
Collaboration and interactions	1	The student discusses and negotiates learning goals or group tasks during class/gamified sessions.	0.915
	2	The student expresses personal needs or emotions regarding the learning process, peers, or activities.	0.884
	3	The student connects current classroom experiences to personal learning patterns or past group events.	0.853
	4	The student introduces meaningful new ideas or topics related to learning goals.	0.911
	5	The student shares their thoughts and feelings about learning in a natural, unprompted way.	0.869
	6	The student links ideas/themes across tasks or lessons (e.g., gamified challenges or levels).	0.898
Active learning	1	I asked questions or contributed to discussions in class or online.	0.918
	2	I made a class or online presentation.	0.900
	3	I worked with other students on projects during class.	0.901
	4	I worked with other students outside class to prepare assignments.	0.907
	5	I tutored or taught other university students (paid or voluntary).	0.914
	6	I participated in a community-based project (e.g., volunteering) as part of study.	0.938
	7	I discussed ideas from readings or classes with others outside class.	0.902
	8	I communicated or worked with students online (e.g., forums, group chats).	0.883
	9	I searched online for resources relevant to my studies.	0.881
	10	I obtained timely online feedback from assessments.	0.856
	11	I persisted with challenging learning activities despite initial setbacks.	0.892
Student aca-demic performance	1	I complete my classwork consistently.	0.865
	3	I complete my classwork accurately.	0.852
	6	I follow instructions during group work.	0.883
	8	I learn new material quickly.	0.776
	11	I speak clearly and confidently during activities.	0.902
	13	I take a reasonable amount of time to finish work.	0.879
	14	I stay focused on tasks without reminders.	0.865
	19	I avoid interacting with others during class.	0.880

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